



Content Processing & Clustering Web Science

COMPSCI4077/COMPSCI5107/COMPSCI5078

Joemon M Jose

Overall info

announcements

Communication

- Through Teams Page
 - so please sign in -
 - [WEB SCIENCE - COMPSCI4077, COMPSCI5078 & COMPSCI5107 - 2024-25 | General | Microsoft Teams](#)
 - If you are not on Teams Channel
 - you will not get any information/emails
- Lecture materials
 - Moodle
 - <https://moodle.gla.ac.uk/course/view.php?id=44995>
 - Ask teaching office to add you, if you are not on moodle
 - Coursework & deadlines
 - Announcement on Week 2

Which module to enroll?

- COMPSCI4077 - Web Science H – For level H students
- COMPSCI5107 - Web Science for MSc
- COMPSCI5078 - Web Science M – For Level M students
- Please enrol for the right module
 - Otherwise affect your degree!!!

Please watch the Introduction lecture

- If you have any questions, please write your questions at
 - https://glamys.sharepoint.com/personal/joemon_jose_glasgow_ac_uk/_layouts/15/doc.aspx?sourcedoc={9185079d-37bf-48dd-84db-aa34e6aeb910}&action=edit
 - I will answer them in the class
 - This will help the rest of the class as well
 - they will get an opportunity to clarify issues

Previous lecture

- Social media, data,
- Social sensors
- Social dynamics
- Tweet Structure
 - components

← **Post**



Claudia Hauff    
@CharlotteHase

It is this time of the year when you should think about early career researchers in the Information Retrieval field that deserve recognition! SIGIR has three categories of awards for early career researchers! sigir.org/awards/sigir-e... Self-nominations are also welcome! By Feb 16

10:34 AM · Dec 18, 2024 · **142** Views

Content processing

Processing Tweet text!

Fo

1615599288544788480.json

```
114 },
115 "geo": null,
116 "coordinates": null,
117 "place": null,
118 "contributors": null,
119 "is_quote_status": false,
120 "extended_tweet": {
121   "full_text": "When tories sold off Royal Mail 16 priority investors were created. One al
2nd biggest went to BlackRock who made up to £27m &amp; later employed Osborne as adviser on £650K a
122   "display_text_range": [
123     0,
124     282
125   ],
126   "entities": {
127     "hashtags": [],
128     "urls": [],
129     "user_mentions": [],
130     "symbols": []

```

```
1 {
2   "created_at": "Wed Jan 18 06:37:41 +0000 2023",
3   "id": 1615599288544788480,
4   "id_str": "1615599288544788480",
5   "text": "RT @withorpe: When tories sold off Royal Mail 16 priority investors were created. One allocated most shares was Capital Group, employer of..."
6   "source": "<a href='\"http://twitter.com/download/android\"' rel='\"nofollow\"'>Twitter for Android</a>",
7   "truncated": false,
8   "in_reply_to_status_id": null,
9   "in_reply_to_status_id_str": null,
10  "in_reply_to_user_id": null,
11  "in_reply_to_user_id_str": null,
12  "in_reply_to_screen_name": null,
13  "user": {
14    "id": 1161317586,
15    "id_str": "1161317586",
16    "name": "Nathan Fry",
17    "screen_name": "NathanFryLeeds",
18    "location": null,
19    "url": null,
20    "description": null,
21    "translator_type": "none",
22    "protected": false,
23    "verified": false,
24    "verified_type": "blue",
25    "followers_count": 119,
26    "friends_count": 299,
27    "listed_count": 1,
28    "favourites_count": 49023,

```

Content - ["text"]

- NO 7. IN THE UK COME ON BOOST IT TO NO 1. WE CAN DO IT 🌟❤️
- 🆚 (Nottm) Nottingham Forest vs (Wolves) Wolverhampton Wanderers 🆚...
<https://t.co/k08F3QsQPA>
- 🚑 Stones has not trained with the #ManCity squad this week. Hearing that the injury is not major but he is a definit...
<https://t.co/HyheG0p00l> You say?????????😬😬😬

Removing stuff

Elites who work behind the scenes

Who contribute time and money

To every single Elite who makes Erica feel special and loved

God bless you 🥺🥺😘❤️ #AkwaabaEricaNlewedim

Non-ascii removal

```
def remove(x, replacer):
    for i in x:
        if f"'{i}'" == ascii(i):
            pass
        else:
            x=x.replace(i,replacer)
    return x
```

-
- educated?""Yes, cowardice
racing, Aloud Nobly.
Contemplate
- ؟؟
- نمشے
- Educated Yes, cowardice
racing, Aloud Nobly.
Contemplate
- ◆H80◆
- ◆H80◆
- ◆H80◆
- H80 H80 H80

Two other cases



- **1st Case**

@gavincost @Freedom16356531 I mean he's not wrong about the alarming hairline 😂

- @gavincost @Freedom16356531 I mean **he s** not wrong about the alarming hairline

- **2nd case**

📺 Live Stream

🔴 PSG vs Angers

🔗 Link1: <https://t.co/0DLHRoDz1q>

🔴 Southampton vs Manchester City

🔗 Link 2 : <https://t.co/ZMyGEloSc7>

🔴 Nottingham Forest vs Wolves

🔗 Link 3 : <https://t.co/902lZ17u8b>

PSGSCO #SOUMCI #ManCity #NFOWOL #NFFC

wk

- [Live Stream](#) [PSG vs Angers](#) [Link1: https://t.co/0DLHRoDz1q](#) [Southampton vs Manchester City](#) [Link 2 : https://t.co/ZMyGEloSc7](#) [Nottingham Forest vs Wolves](#) [Link 3 : https://t.co/902lZ17u8b](#) # [#PSGSCO](#) [#SOUMCI](#) [#ManCity](#) [#NFOWOL](#) [#NFFC](#) wk

Next step

- Once the data is cleaned
 - *We need to process them*
 - *For identifying important tweets*
 - *Events*
 - *People reactions*
- As a first step we will focus on grouping similar tweets

Tweet text – content processing

- {'_id': '1351840236154867712', 'date': 'Wed Jan 20 10:33:22 +0000 2021', 'username': 'Surailian', 'text': '***Data from the vaccination programme in Israel raises serious questions about the UK government's massive gamble to extend the gap between #COVID19 shots. Watch the whole clip to understand the issue. Once you have, shout about it. We can't let the Tories blow our only way out'***'}
- **Tweets – text**
- Data from the vaccination programme in Israel raises serious questions about the UK government's massive gamble to extend the gap between [#COVID19](#) shots. Watch the whole clip to understand the issue. Once you have, shout about it. We can't let the Tories blow our only way out.
-

tokenization

- Original text

- @Princess_Kit_K I suppose they re the two main liability issues, theft and safety, from an insurance standpoint? Particularly in older cars. I suspect we re similar ages. I like the fact the US has the firemen/paramedics dual trained, seems incredibly useful. No crossover in the UK, due to NHS.

- **Vector representation**

- ['@Princess_Kit_K', 'ages', 'cars', 'crossover', 'dual', 'due', 'fact', 'firemenparamedics', 'incredibly', 'insurance', 'issues', 'liability', 'like', 'main', 'nhs', 'older', 'particularly', 'safety', 'seems', 'similar', 'standpoint', 'suppose', 'suspect', 'theft', 'trained', 'two', 'useful']
- **Some tokens missing!, why?**

What is done?

- I m selling 2x tickets for Manchester United Vs Manchester City Seated together for 14th January Message me if you re interested for proofs#ManCity

- Separate each token

- Remove stopwords – what are they?

- I m 2x for Vs me if you re

```
['14th', 'city', 'interested', 'january',  
'manchester', 'message', 'proofsmancity',  
'seated', 'selling', 'tickets', 'together',  
'united'],
```

Another example

- Text

- @MarioNawfal And she already went to the police and yea, they didn t do anything about it because of the lack of evidence or something else, and I don t know how England goes with out of date cases. because 10 years is a long time, and he was a nobody, yea a kickboxer.

- Corresponding vector

- ['@MarioNawfal', 'already', 'anything', 'casesbecause', 'date', 'elseand', 'england', 'evidence', 'goes', 'kickboxer', 'know', 'lack', 'long', 'nobody', 'police', 'something', 'time', 'went', 'yea', 'years']

Another example

- Text

- @MarioNawfal And she already went to the police and yea, they didn't do anything about it because of the lack of evidence or something else, and I don't know how England goes with out of date cases.because 10 years is a long time, and he was a nobody, yea a kickboxer.

- Corresponding vector

- ['@MarioNawfal', 'already', 'anything', 'casesbecause', 'date', 'elseand', 'england', 'evidence', 'goes', 'kickboxer', 'know', 'lack', 'long', 'nobody', 'police', 'something', 'time', 'went', 'yea', 'years']

Vector Representation

- Documents are represented by a term vector

$$D_i = (t_{i1}, t_{i2}, \dots, t_{in})$$

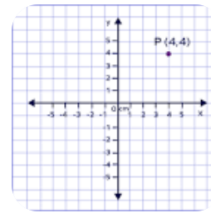
- Queries are represented by a similar vector

$$Q = (qt_1, qt_2, \dots, qt_n)$$

- In binary scheme, t_{ik} (or q_k) is set to 1 when term k is present in Document i (or in query) otherwise they are set to zero

◆ AI Overview

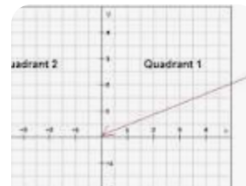
In algebra, a coordinate system is **a two-dimensional number line with two perpendicular axes, the x-axis and the y-axis, that intersect at the origin**. The point where the axes meet is the origin, and its coordinates are $(0, 0)$. [🔗](#)



A point on the coordinate plane is described by an ordered pair of numbers, (x, y) , where the first number is the x-coordinate and the second is the y-coordinate. The x-coordinate represents the distance along the horizontal axis, and the y-coordinate represents the distance along the vertical axis. [🔗](#)

Show more [▼](#)

A coordinate system is **a two-dimensional number line**, for example, two perpendicular number lines or axes. The center of the coordinate system (where the lines intersect) is called the origin. The axes intersect when



SplashLearn



Coord
Intro

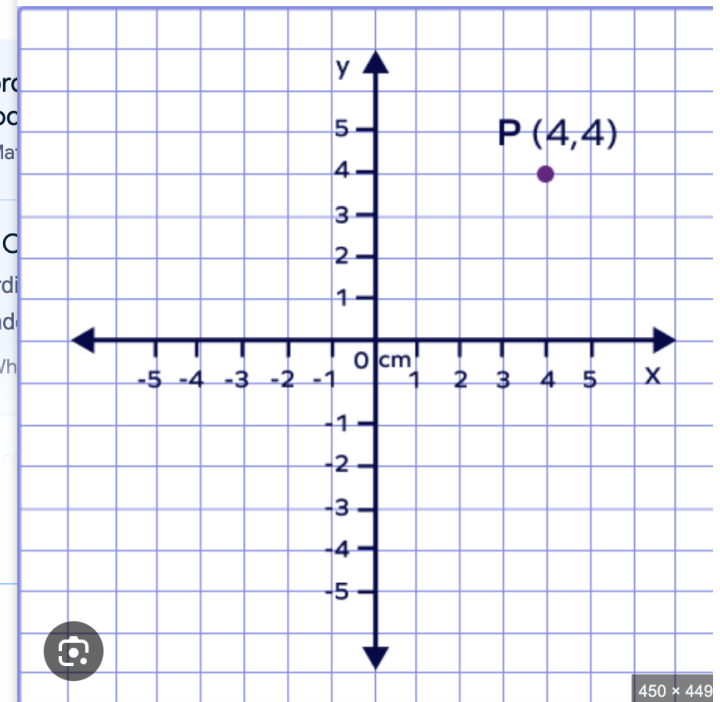
Ma

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to und

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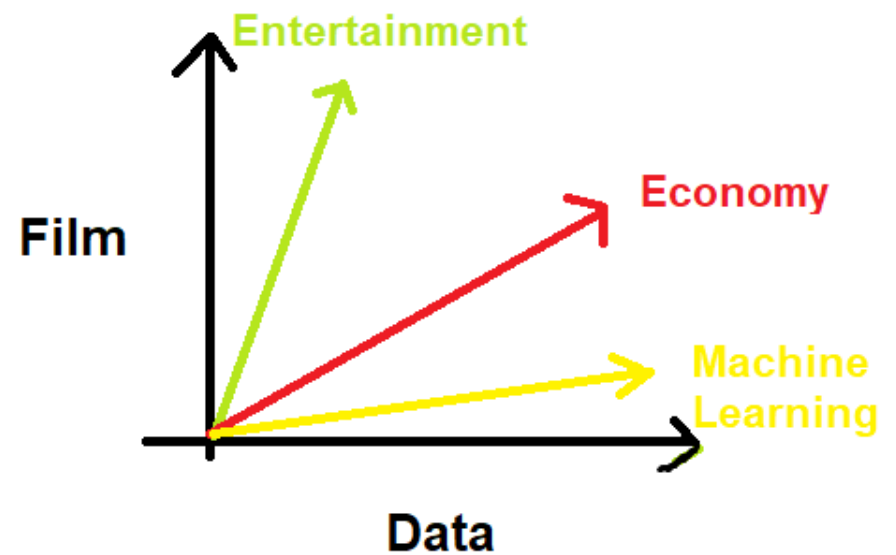
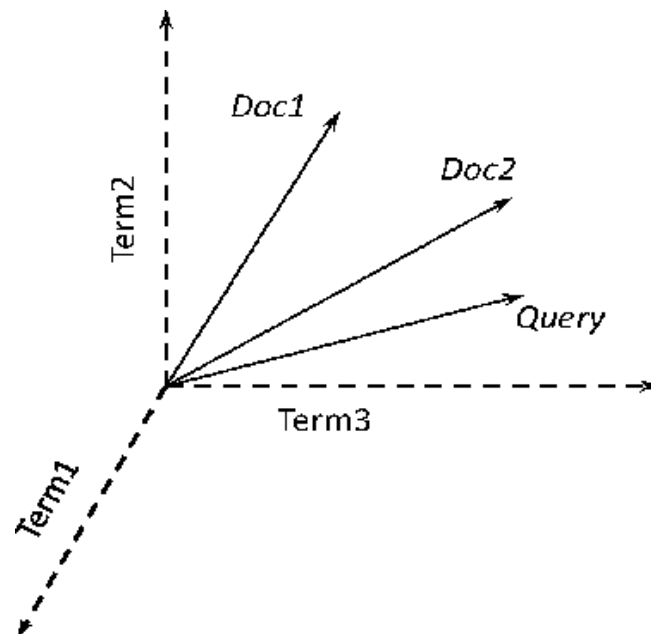
450 x 449

What are Coordinates in Math?
Definition, Examples, Facts

Visit >

Visualising

- Vector-space representation



Similar documents

- The most relevant documents for a query are expected to be those represented by the vectors closest to the query, that is documents that use similar words to the query.
- Closeness is often calculated by just looking at angles between document vector and query vector
- We need a similarity measure!

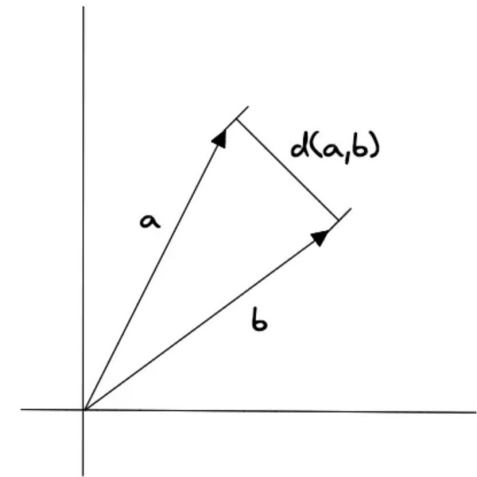
Cosine Similarity Measure

$$D = (t_1, t_2, t_3, \dots, t_n)$$

$$Q = (qt_1, qt_2, qt_3, \dots, qt_n)$$

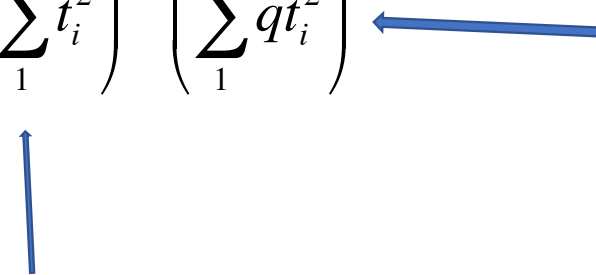
$$\cos(D, Q) = \frac{\sum_{i=1}^n t_i * qt_i}{\left(\sum_{i=1}^n t_i^2 \right)^{1/2} \left(\sum_{i=1}^n qt_i^2 \right)^{1/2}}$$

$$\begin{aligned}\cos(0) &= 1 \\ \cos(90) &= 0\end{aligned}$$

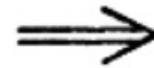


Length of a vector

- The magnitude of a vector is **its size**.
 - It can be calculated from the square root of the sum of the squares of the individual vector components.

$$\cos(D, Q) = \frac{\sum_{i=1}^n t_i * qt_i}{\left(\sum_{i=1}^n t_i^2 \right)^{1/2} \left(\sum_{i=1}^n qt_i^2 \right)^{1/2}}$$


this vector
has a length
of 10



this vector has
a length of 1



the process of
normalization

862 × 248

What is a normalized (unit) vector?



unit vector

- To normalize a vector, therefore, is to take a vector of any length and, keeping it pointing in the same direction, change its length to 1, turning it into what is called a **unit vector**.
- to normalize a vector, simply divide each component by its magnitude.
- The magnitude of a vector is **its size**.
 - It can be calculated from the square root of the sum of the squares of the individual vector components.
- Suppose each term is weighted by 1
 - Then if we divide by the sqrt of the total number of components, we should get a normalised weight.
 - If for example a tweet vector has 19 terms – we give a weight of $1/\text{Sqrt}(19)$

For any given vector $\vec{u}$, its unit vector (written as $\hat{u}$) is calculated as follows:

$$\hat{u} = \frac{\vec{u}}{||\vec{u}||}$$

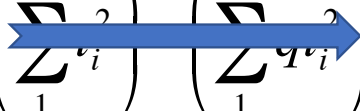
Similarity computation

- ~~Dice coefficient~~
- ~~Jaccard coefficient~~
- Cosine similarity
 - 0...1
- Similarity vs dissimilarity (Distance)
 - Distance = 1 - similarity
- Working with normalized vectors
 - Benefits are ...
 - Similarity can be computed just using
 - Sum of $t_i * qt_i$

Cosine Similarity Measure – benefits of a normalised vector

$$D = (t_1, t_2, t_3, \dots, t_n)$$

$$Q = (qt_1, qt_2, qt_3, \dots, qt_n)$$

$$\cos(D, Q) = \frac{\sum_{i=1}^n t_i * qt_i}{\left(\sum_{i=1}^n t_i^2 \right)^{1/2} \left(\sum_{i=1}^n qt_i^2 \right)^{1/2}}$$


So the idea is to normalise after each step!

Vector representation

- @Fletch19852 @philcooper15 @Lee_Smithh
@SkySportsNews @swiddoy76 So why are you trying
so hard to defend a concrete stand. It s in
Sheffield Wednesdays fans culture to live in
denial. To make up lies.
- ['@Fletch19852', '@Lee_Smithh',
'@SkySportsNews', '@philcooper15',
'@swiddoy76', 'concrete', 'culture', 'defend',
'denial', 'fans', 'hard', 'lies', 'live',
'sheffield', 'stand', 'trying', 'wednesdays'],

Tokenize & normalize the string!

- if(token.startswith('#') or token.startswith('@') or token.startswith('\$')):

return token

- if(token.startswith('T&') or token.startswith('https')):
return None

- *s = token.lower()*
 - *exclude = set(string.punctuation)*

s = ''.join(ch for ch in s if ch not in exclude)

- @Princess_Kit_K I suppose they re the two main liability issues, theft and safety, from an insurance standpoint? Particularly in older cars. I suspect we re similar ages. I like the fact the US has the firemen/paramedics dual trained, seems incredibly useful. No crossover in the UK, due to NHS.
- ['@Princess_Kit_K', 'ages', 'cars', 'crossover', 'dual', 'due', 'fact', 'firemenparamedics', 'incredibly', 'insurance', 'issues', 'liability', 'like', 'main', 'nhs', 'older', 'particularly', 'safety', 'seems', 'similar', 'standpoint', 'suppose', 'suspect', 'theft', 'trained', 'two', 'useful']

-
- @Princess Kit_K I suppose they re the two main liability issues, theft and safety, from an insurance standpoint? Particularly in older cars. I suspect we re similar ages. I like the fact the US has the firemen/paramedics dual trained, seems incredibly useful. No crossover in the UK, due to NHS.
 - ['@Princess_Kit_K', 'ages', 'cars', 'crossover', 'dual', 'due', 'fact', 'firemenparamedics', 'incredibly', 'insurance', 'issues', 'liability', 'like', 'main', 'nhs', 'older', 'particularly', 'safety', 'seems', 'similar', 'standpoint', 'suppose', 'suspect', 'theft', 'trained', 'two', 'useful']

```
def remove(x, replacer):  
    for i in x:  
        if f"{'{i}'}" == ascii(i):  
            pass  
        else:  
            x=x.replace(i,replacer)  
    return x
```

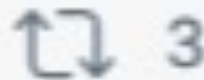
So far last week

- Vectorisation of Tweet Text
 - In other words, **we are representing each tweet**
 - We will use it for finding similarity, ...
- Focused on the tweet text
 - Removed non-ascii
 - Tokenise the text string - Split tokens
 - Normalise the tokens
 - Removed stopwords
 - Stopword file + situation specific
 - Remaining terms are our features
 - Create a vector representation
- Text document
 - We talked about vector spaces
 - A mathematical concept
 - We apply that to our term space
 - Unique terms as axes
- Normalise the vector
 - Create a unit vector



Vocations A and B @ABatVocations1 · Jan 21, 2019

Memorial of St. **Agnes**, virgin and martyr; Our Lady o
(**Dominican** Republic) - **January** 21, 2019 - Liturgica
Culture catholicculture.org/culture/liturg...



Vector
representation

- ['2021', 'agnes', 'calendar', 'catholic', 'culture', 'dominican', 'grace', 'high', 'https://t.co/IJnKzDfhUs', 'january', 'lady', 'liturgical', 'martyr', 'memorial', 'republic', 'virgin']

Term weighting

- Assigning importance to unique words
 - In other word features
 - How important each dimension is?
- In documents, as in web pages
 - If a word is repeated many times,
 - it means that the document is about that word!
 - Repetition means importance
 - Term frequency?
- However, in tweets
 - Often words are not repeated – short text
 - So, we assume all words are equally important
 - If a word is present, then that dimension is important in our underlying vector representation

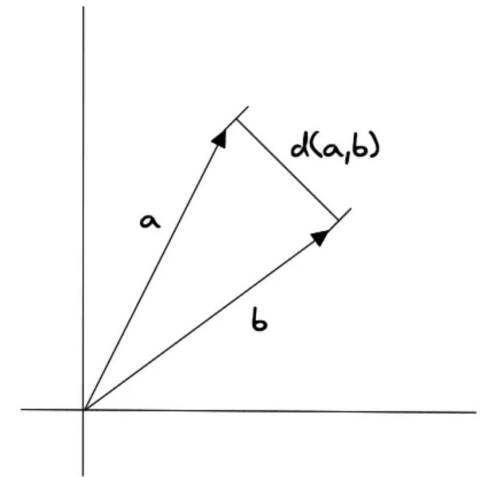
Cosine Similarity Measure

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$$\begin{aligned}\cos(0) &= 1 \\ \cos(90) &= 0\end{aligned}$$



Vector representation - weighting

- ['2021', 'agnes', 'calendar', 'catholic', 'culture', 'dominican', 'grace', 'high', 'https://t.co/IJnKzDfhUs', 'january', 'lady', 'liturgical', 'martyr', 'memorial', 'republic', 'virgin']
- Two options
 - 16 elements
 - An Ad-hoc mechanism
 - 1 each
 - [1,1,1, ...,1]
 - [0,0,0,1,1,0, ...,0,0, ...,1]
 - Or
 - Assuming all terms appear once, we can assign a weight
 - 1/16
 - [.06,.06, .06, ..., .06]
 - [0,0,0, .06, .06,0, ...,0,0, ..., .06]
 - Then normalize!

So far!

- Focused on the tweet text
- Removed non-ascii
- Tokenise the text string
- Normalise the tokens
- Create a vector representation
- Normalise the vector
 - Create unit vector



Finding similar tweets

Grouping Tweets - Single-pass clustering

What we have

- Large stream of tweets
- Very Short in length – 280 characters +
- Noisy in nature
- ***So, the processing should be***
 - ***Fast***
 - ***robust***

What is a cluster?

- **In General Usage**

- A **cluster** is simply a collection or group of similar things located close together.
 - Example: A cluster of grapes, buildings, or ideas.

- **In our context**

- group of similar tweets grouped together based on certain characteristics or features.
- Common words / features

Single-pass clustering

- Requires a single, sequential pass over the set of documents it attempts to cluster.
- The idea is to group similar documents (tweets) as and when it arrives
 - Sequential nature
- The algorithm classifies a document (the next one) in the sequence
 - based on the current set of clusters and
 - according to a condition on the similarity function employed.

Single-pass clustering ...

- At every stage, the algorithm decides on whether a newly seen document should become a member of an already defined cluster or the centre of a new one.
 - In its most simple form, the similarity function gets defined based on
 - just some similarity (or alternatively, dissimilarity) measure
 - between document-feature vectors.
- Cosine Similarity

Cluster centroid

- Given a set of documents (tweets)
 - We take the average of the vector representation
 - As a representation of the cluster/group
- Remember, we are grouping based on the content representation
 - Hence, we assume the documents are similar
 - In terms of word similarity and hence
 - Semantic similarity!
- Cluster centroid is the representation of the group/cluster
 - Assume it as one document

Example cluster centroid

- ```
[('denzel', 0.31507569531533086), ('dumfries', 0.31507569531533086), ('inter', 0.31507569531533086), ('ahead', 0.2171929849239275), ('building', 0.2171929849239275), ('clash', 0.2171929849239275), ('italiana', 0.2171929849239275), ('milan', 0.2171929849239275), ('photo', 0.2171929849239275), ('supercoppa', 0.2171929849239275), ('wednesday', 0.2171929849239275), ('wingback', 0.2171929849239275), ('#ForzaInter', 0.11988134754977947), ('#Internazionale', 0.11988134754977947), ('#Internisti', 0.11988134754977947), ('#Nerazzuri', 0.11988134754977947), ('#COYS', 0.0978827103914034), ('#THFC', 0.0978827103914034), ('23yearold', 0.0978827103914034), ('26yearold', 0.0978827103914034), ('@SempreIntercom', 0.0978827103914034), ('antonio', 0.0978827103914034), ('better', 0.0978827103914034), ('characteristics', 0.0978827103914034), ('conte', 0.0978827103914034), ('decided', 0.0978827103914034), ('interested', 0.0978827103914034), ('pedro', 0.0978827103914034), ('player', 0.0978827103914034), ('porro', 0.0978827103914034), ('prefer', 0.0978827103914034), ('rightback', 0.0978827103914034), ('sporting', 0.0978827103914034), ('spurs', 0.0978827103914034), ('technical', 0.0978827103914034), ('tier', 0.0978827103914034), ('wants', 0.0978827103914034), ('#FCIM', 0.09731163737414802), ('#InterMilan', 0.09731163737414802)]
```
- ```
***** the tweets in the cluster *****
```
- ```
1 - 1615589931195187204 InsideSpurs1 :-- : Spurs are NOT interested in 26-year-old Inter right-back Denzel Dumfries and prefer Sporting s 23-year-old Pedro Porro. Antonio Conte has decided he wants a player with better technical characteristics than Dumfries. @SempreIntercom - Tier #THFC || #COYS https://t.co/aI8NEBz5k8 qScore : 0.6179394977168949 nScore : 4.4073424851295515
```
- ```
2 - 1615591069046939650 InterMilanNow :-- : Photo Inter Milan Wingback Denzel Dumfries Ahead Of Supercoppa Italiana Clash Vs AC Milan: Building Up To Wednesday #ForzaInter #Internisti #Nerazzuri #Internazionale https://t.co/j69E8iq094 qScore : 0.6718820861678004 nScore : 1.9407670315888352
```
- ```
3 - 1615592490526023680 intermilan :-- : Photo - Inter Milan Wingback Denzel Dumfries Ahead Of Supercoppa Italiana Clash Vs AC Milan: "Building Up To Wednesday" #InterMilan #FCIM https://t.co/DBiPW8BvCN qScore : 0.6643733850129199 nScore : 2.8329581689327106
```
- ```
-----
```

Stream of tweets

- 100 Million tweets a day
 - Tweets arrive one after another
- We want to cluster them as they arrive

Single Pass Clustering

- The general algorithm is as follows:
- **Step 1.**
 - Assign the first document D_1 as the representative for C_1 .
 - Document vector becomes the **cluster vector/centroid**
- **Step 2.**
 - For each D_i , calculate the similarity S with the representative for each existing clusters.
 - Document vector D_i is compared to **each cluster centroids**
 - **We need a cluster centroid – or an average representation of documents in the cluster**
 - Let us say, S_{max} is the maximum similarity of the incoming document and existing clusters
- **Step 3.**
 - If S_{max} is greater than a threshold value S_T ,
 - add the item to the corresponding cluster  We need to make sure the similarity is substantial
 - and recalculate the cluster representative;
 - otherwise, use D_i to initiate a new cluster.
- **Step 4.**
 - For a new item D_i to be clustered, return to step 2.

How to group tweets

Parameters

Cosine similarity
Similarity Threshold

```
• maxSim=0
• sim =0
• if (self.clusterCounter ==0):
•     self.createCluster(id, text, self.clusterCounter, time)
•     self.clusterCounter += 1 # update ClusterCounter
• else:
•     for cluster in self.clusterList:
•         sim = cluster.computeSimilarity(text)
•         # print(sim, maxSim)
•         if(sim > maxSim):
•             maxSim = sim
•             cid = cluster.clusterId
•     # out of for loop
•     # threshold similarity 2
```

First doc form a cluster

1) **for each document d in the sequence loop**
a) find a cluster c that maximises $\cos(c, d)$;
b) **if** $\cos(c, d) > t$ **then** include d in c ;
c) **else** create a new cluster whose only document is d ;
2) **end loop.**

```
→ if (maxSim < 0.2):
•     self.createCluster(id, text, self.clusterCounter, time)
•     self.clusterCounter += 1 # update ClusterCounter

• else:
•     clusterN = self.clusterList[cid]
•     clusterN.add(id, text, time)
•     # self.clusterList[cid].printCluster()

• except Exception as e:
•     print(e)
```

You check the similarity above a threshold, to make sure both document are similar

Otherwise you create a new cluster

Comments

- The single pass method is particularly simple (and useful in this scenario)
 - since it requires that the data set be processed only once
 - and as and when it arrives
- Obviously, the results for this method are highly dependent
 - on the similarity threshold that is used.

Comments & Observations

- It tends to produce large clusters early in the clustering pass,
 - and because the clusters formed are not independent of the order in which the data set is processed,
 - Because for similar documents but in different order would have created different cluster
 - You should use **your judgment in setting this threshold** so that you are left with a reasonable number of clusters.
- Post-processing tips!
 - Use to form the groups initially, which are used to reallocate/create new clusters.
 - If we get a large noisy clusters of tweets, we could re-cluster them!!!

Single pass clustering

- <http://facweb.cs.depaul.edu/mobasher/classes/csc575/assignments/single-pass.html>
- Let us look at some cluster behaviours

Cluster centroids

the cluster vector/centroid

- Wed Jan 20 10:47:14 +0000 2021
- Wed Jan 20 10:48:15 +0000 2021
- 8
- { '@SenSherrrodBrown': 0.0001293425917369981, 'dems': 0.0001293425917369981, 'flee': 0.0001293425917369981, 'friends': 0.0001293425917369981, 'good': 0.0005173703669479924, 'high': 0.0001293425917369981, 'lot': 0.0001293425917369981, 'moved': 0.0001293425917369981, 'new': 0.2379813708804682, 'notion': 0.0001293425917369981, 'pay': 0.0001293425917369981, 'people': 0.0001293425917369981, 'real': 0.0001293425917369981, 'scoff': 0.0001293425917369981, 'taxes': 0.0001293425917369981, 'york': 0.2379813708804682, '@JackObinyan': 0.0005173703669479924, 'ambassadorship': 0.0005173703669479924, 'capable': 0.0005173703669479924, 'handling': 0.0005173703669479924, 'woman': 0.0005173703669479924, '@jnonpark': 0.0015358581807091989, 'englishman': 0.0015358581807091989, '2015': 0.003355437015213525, 'footage': 0.003355437015213525, 'https://t.co/CwWD8AfShR': 0.003355437015213525, 'irving': 0.003355437015213525, 'plaza': 0.003355437015213525, 'century': 0.009238719597542596, 'chowder': 0.009238719597542596, 'comes': 0.009238719597542596, 'often': 0.009238719597542596, 'reading': 0.009238719597542596, 'surprised': 0.009238719597542596, 'turn': 0.009238719597542596, '@brownboyrod': 0.028827443225059805, 'apartment': 0.028827443225059805, 'buy': 0.028827443225059805, 'still': 0.028827443225059805, 'waiting': 0.028827443225059805, 'ass': 0.08048269368968919, 'federal': 0.08048269368968919, 'look': 0.08048269368968919, 'president': 0.2379813708804682, 'prison': 0.08048269368968919, 'time': 0.08048269368968919, '@StormisHereNow': 0.2379813708804682, 'aimed': 0.2379813708804682, 'anyone': 0.2379813708804682, 'dallas': 0.2379813708804682, 'incoming': 0.2379813708804682, 'knowthe': 0.2379813708804682, 'myyou': 0.2379813708804682, 'nuclear': 0.2379813708804682, 'seen': 0.2379813708804682, 'sir': 0.2379813708804682, 'thing': 0.2379813708804682, 'want': 0.2379813708804682, 'warheads': 0.2379813708804682, 'washington': 0.2379813708804682 }
- 1351843722087059457
- 1351843736620249089
- 1351843757813952514h
- 1351843829230555136
- 1351843852286627841
- 1351843873580990464
- 1351843894812696577
- 1351843981374746629
- ['@SenSherrrodBrown', 'dems', 'flee', 'friends', 'good', 'high', 'lot', 'moved', 'new', 'notion', 'pay', 'people', 'real', 'scoff', 'taxes', 'york']
- ['@JackObinyan', 'ambassadorship', 'capable', 'good', 'handling', 'new', 'woman', 'york']
- ['@jnonpark', 'englishman', 'new', 'york']
- ['2015', 'footage', 'https://t.co/CwWD8AfShR', 'irving', 'new', 'plaza', 'york']
- ['century', 'chowder', 'comes', 'new', 'often', 'reading', 'surprised', 'turn', 'york']
- ['@brownboyrod', 'apartment', 'buy', 'new', 'still', 'waiting', 'york']
- ['ass', 'federal', 'look', 'new', 'president', 'prison', 'time', 'york']
- ['@StormisHereNow', 'aimed', 'anyone', 'dallas', 'incoming', 'knowthe', 'myyou', 'new', 'nuclear', 'president', 'seen', 'sir', 'thing', 'want', 'warheads', 'washington', 'york']
- ***** end cluster print *****

How could we create this cluster centroid?

Group representation – average vector!

- {'@SenSherrodBrown': 0.0001293425917369981, 'dems': 0.0001293425917369981, 'flee': 0.0001293425917369981, 'friends': 0.0001293425917369981, 'good': 0.0005173703669479924, 'high': 0.0001293425917369981, 'lot': 0.0001293425917369981, 'moved': 0.0001293425917369981, 'new': 0.2379813708804682, 'notion': 0.0001293425917369981, 'pay': 0.0001293425917369981, 'people': 0.0001293425917369981, 'real': 0.0001293425917369981, 'scoff': 0.0001293425917369981, 'taxes': 0.0001293425917369981, 'york': 0.2379813708804682, '@JackObinyan': 0.0005173703669479924, 'ambassadorship': 0.0005173703669479924, 'capable': 0.0005173703669479924, 'handling': 0.0005173703669479924, 'woman': 0.0005173703669479924, '@jnonpark': 0.0015358581807091989, 'englishman': 0.0015358581807091989, '2015': 0.003355437015213525, 'footage': 0.003355437015213525, 'https://t.co/CwWD8Af5hR': 0.003355437015213525, 'irving': 0.003355437015213525, 'plaza': 0.003355437015213525, 'century': 0.009238719597542596, 'chowder': 0.009238719597542596, 'comes': 0.009238719597542596, 'often': 0.009238719597542596, 'reading': 0.009238719597542596, 'surprised': 0.009238719597542596, 'turn': 0.009238719597542596, '@brownboyrod': 0.028827443225059805, 'apartment': 0.028827443225059805, 'buy': 0.028827443225059805, 'still': 0.028827443225059805, 'waiting': 0.028827443225059805, 'ass': 0.08048269368968919, 'federal': 0.08048269368968919, 'look': 0.08048269368968919, 'president': 0.2379813708804682, 'prison': 0.08048269368968919, 'time': 0.08048269368968919, '@StormisHereNow': 0.2379813708804682, 'aimed': 0.2379813708804682, 'anyone': 0.2379813708804682, 'dallas': 0.2379813708804682, 'incoming': 0.2379813708804682, 'knowthe': 0.2379813708804682, 'myyou': 0.2379813708804682, 'nuclear': 0.2379813708804682, 'seen': 0.2379813708804682, 'sir': 0.2379813708804682, 'thing': 0.2379813708804682, 'want': 0.2379813708804682, 'warheads': 0.2379813708804682, 'washington': 0.2379813708804682}

Computation

- We need to store the tweet vectors along with weights
 - ['@SenSherrodBrown', 'dems', 'flee', 'friends', 'good', 'high', 'lot', 'moved', 'new', 'notion', 'pay', 'people', 'real', 'scoff', 'taxes', 'york']
 - ['@JackObinyan', 'ambassadorship', 'capable', 'good', 'handling', 'new', 'woman', 'york']
 - ['@jnonpark', 'englishman', 'new', 'york']
 - ['2015', 'footage', 'https://t.co/CwWD8Af5hR', 'irving', 'new', 'plaza', 'york']
 - ['century', 'chowder', 'comes', 'new', 'often', 'reading', 'surprised', 'turn', 'york']
 - ['@brownboyrod', 'apartment', 'buy', 'new', 'still', 'waiting', 'york']
 - ['ass', 'federal', 'look', 'new', 'president', 'prison', 'time', 'york']
 - ['@StormisHereNow', 'aimed', 'anyone', 'dallas', 'incoming', 'knowthe', 'myyou', 'new', 'nuclear', 'president', 'seen', 'sir', 'thing', 'want', 'warheads', 'washington', 'york']
- *Compute the average of those vectors*
- *Issues*
 - *Storage*
 - *computation*

Average vector/cluster centroid - efficiency

- If we have a collection of **n** documents in the same cluster/class.
- if the feature vectors for these examples are
 - $\{ x(1), x(2), \dots, x(n) \}$,
 - the mean vector m is by averaging:
 - $m(n) = [x(1) + x(2) + \dots + x(n)] / n$.
- Problems with this approach ...
 - Not very efficient

moving average

- $m(n) = [x(1) + x(2) + \dots + x(n)] / n$.
- We can use a moving average in streaming cases
 - Let us say, the sum of the first n examples is $n * m(n)$
 - When a new item arrives,
 - the sum of the first $(n+1)$ examples is
 - $n * m(n) + x(n+1)$.
- **Average Vector**
 - **is to divide by $(n+1)$**
 - If we divide by $n+1$, we obtain the following useful recursive formula
 - for sequential accumulation of the mean :

$$m(n + 1) = \frac{n}{n + 1} * m(n) + \frac{1}{n + 1} x(n + 1)$$

Storage

$n, m(n)$ for each cluster

Average vector example

- 1 - 1614729262396735490 hulllive :-- : Our thoughts are with those who have lost a loved one <https://t.co/H9k07TLPM3> qScore : 0.7556686777275012 nScore : 2.0143721579776903
- 2 - 1614729275651018754 EADT24 :-- : I remember seeing an interview with the fabulous Dame Maggie Smith some years ago in which she spoke movingly about the loss of her husband and said that one of the worst things about widowhood was that you no longer felt special to anyone. <https://t.co/yaI3FemtmP> qScore : 0.7279856656371267 nScore : 2.436056117412133
- 3 - 1614729276905132032 ipswichstar24 :-- : I remember seeing an interview with the fabulous Dame Maggie Smith some years ago in which she spoke movingly about the loss of her husband and said that one of the worst things about widowhood was that you no longer felt special to anyone. <https://t.co/Sdan589kR4> qScore : 0.728945810564663 nScore : 2.436056117412133
- 4 - 1614729278129868802 EveningNews :-- : I remember seeing an interview with the fabulous Dame Maggie Smith some years ago in which she spoke movingly about the loss of her husband and said that one of the worst things about widowhood was that you no longer felt special to anyone. <https://t.co/2Q2i2JJICv7> qScore : 0.7310829073388566 nScore : 2.436056117412133
- -----
- [({'lost', 0.5576872948082074), ('loved', 0.5576872948082074), ('thoughts', 0.5576872948082074), ('one', 0.05785959022863435), ('ago', 0.05785959022863435), ('dame', 0.05785959022863435), ('fabulous', 0.05785959022863435), ('felt', 0.05785959022863435), ('husband', 0.05785959022863435), ('interview', 0.05785959022863435), ('longer', 0.05785959022863435), ('loss', 0.05785959022863435), ('maggie', 0.05785959022863435), ('movingly', 0.05785959022863435), ('remember', 0.05785959022863435), ('seeing', 0.05785959022863435), ('smith', 0.05785959022863435), ('special', 0.05785959022863435), ('spoke', 0.05785959022863435), ('things', 0.05785959022863435), ('widowhood', 0.05785959022863435), ('worst', 0.05785959022863435), ('years', 0.05785959022863435)]

Let us look at a cluster of 3 docs

- 1 - **1615262973425074177** amile_buthelezi :-- :
Guys, are there any Uber blacks in east London??
Genuine chats Or a driver with a proper car someone
knows
- 2 - **1615262994241409024** rjc_1983 :-- : @SW_Help
I am booked on the 3.35PM from London Waterloo to
Southampton today. It looks like this is not running.
I see something about landslip. Your travel planner is
giving no alternative. How do I get home today?
- 3 - **1615269612425302019** Kimkjaerside :-- : Good
morning Another chilly one cycling in today and I
only saw two other cyclists, felt like I was cycling
in a Saturday morning Take care and stay warm
#cyclinglife #Edinburgh <https://t.co/>

Let us look at the cluster after the 2nd doc

- ['1615262973425074177', '1615262994241409024']
- Cluster centroid – first add the vectors
- {'blacks': 0.20851441405707474, 'car': 0.20851441405707474, 'chats': 0.20851441405707474, 'driver': 0.20851441405707474, 'east': 0.20851441405707474, 'genuine': 0.20851441405707474, 'guys': 0.20851441405707474, 'knows': 0.20851441405707474, 'london': 0.1702513061517497, 'proper': 0.20851441405707474, 'someone': 0.20851441405707474, 'uber': 0.20851441405707474, '335pm': 0.1702513061517497, '@SW Help': 0.1702513061517497, 'alternative': 0.1702513061517497, 'booked': 0.1702513061517497, 'giving': 0.1702513061517497, 'home': 0.1702513061517497, 'landslip': 0.1702513061517497, 'like': 0.1702513061517497, 'looks': 0.1702513061517497, 'planner': 0.1702513061517497, 'running': 0.1702513061517497, 'see': 0.1702513061517497, 'something': 0.1702513061517497, 'southampton': 0.1702513061517497, 'today': 0.1702513061517497, 'travel': 0.1702513061517497, 'waterloo': 0.1702513061517497}

Moving average with three documents?

- {`'blacks': 0.13900960937138315,` `'car': 0.13900960937138315,`
`'chats': 0.13900960937138315,` `'driver': 0.13900960937138315,`
`'east': 0.13900960937138315,` `'genuine': 0.13900960937138315,`
`'guys': 0.13900960937138315,` `'knows': 0.13900960937138315,`
`'london': 0.11350087076783312,` `'proper': 0.13900960937138315,` `'someone': 0.13900960937138315,` `'uber':`
`0.13900960937138315,` `'335pm': 0.11350087076783312,`
`'@SW_Help': 0.11350087076783312,` `'alternative': 0.11350087076783312,`
`'booked': 0.11350087076783312,`
`'giving': 0.11350087076783312,` `'home': 0.11350087076783312,`
`'landslip': 0.11350087076783312,` `'like': 0.11350087076783312,`
`'looks': 0.11350087076783312,`
`'planner': 0.11350087076783312,` `'running': 0.11350087076783312,`
`'see': 0.11350087076783312,`
`'something': 0.11350087076783312,` `'southampton': 0.11350087076783312,` `'travel':`
`0.11350087076783312,` `'today': 0.11350087076783312,` `'waterloo': 0.11350087076783312}`

```
for k1, v1 in self.avgVector.items():
    self.avgVector[k1] = (self.numberOfDocuments/(self.numberOfDocuments + 1))*v1
```

Let us look, after 2nd doc

'blacks': 0.20851441405707474,

Reweighting after the third
Document added : so far n=2

2/3 * 0.20851441405707474

'blacks': 0.13900960937138315,

• {'blacks': 0.13900960937138315,
'car': 0.13900960937138315, 'chats':
0.13900960937138315, 'driver': 0.13900960937138315,
'east': 0.13900960937138315, 'genuine':
0.13900960937138315, 'guys': 0.13900960937138315,
'knows': 0.13900960937138315, 'london':
0.11350087076783312, 'proper': 0.13900960937138315,
'someone': 0.13900960937138315, 'uber':
0.13900960937138315, '335pm': 0.11350087076783312,
'@SW_Help': 0.11350087076783312, 'alternative':
0.11350087076783312, 'booked': 0.11350087076783312,
'giving': 0.11350087076783312, 'home':
0.11350087076783312, 'landslip':
0.11350087076783312, 'like': 0.11350087076783312,
'looks': 0.11350087076783312, 'planner':
0.11350087076783312, 'running': 0.11350087076783312,
'see': 0.11350087076783312, 'something':
0.11350087076783312, 'southampton':
0.11350087076783312, 'today': 0.11350087076783312,
'travel': 0.11350087076783312, 'waterloo':
0.11350087076783312}

Reweighting the terms

$$m(n + 1) = \frac{n}{n + 1} * m(n) + \frac{1}{n + 1} x(n + 1)$$

- 1615269612425302019
- ['#Edinburgh', '#cyclinglife', 'another', 'care', 'chilly', 'cycling', 'cyclists', 'felt', 'good', 'like', 'morning', 'one', 'saturday', 'saw', 'stay', 'take', 'today', 'two', 'warm']
- 2023-01-17 08:47:40

How would we weight the terms?

- ['#Edinburgh', '#cyclinglife', 'another', 'care', 'chilly', 'cycling', 'cyclists', 'felt', 'good', 'like', 'morning', 'one', 'saturday', 'saw', 'stay', 'take', 'today', 'two', 'warm']
- {'#Edinburgh': 0.07647191129018724, '#cyclinglife': 0.07647191129018724, 'another': 0.07647191129018724, 'care': 0.07647191129018724, 'chilly': 0.07647191129018724, 'cycling': 0.07647191129018724, 'cyclists': 0.07647191129018724, 'felt': 0.07647191129018724, 'good': 0.07647191129018724, 'like': 0.07647191129018724, 'morning': 0.07647191129018724, 'one': 0.07647191129018724, 'saturday': 0.07647191129018724, 'saw': 0.07647191129018724, 'stay': 0.07647191129018724, 'take': 0.07647191129018724, 'today': 0.07647191129018724, 'two': 0.07647191129018724, 'warm': 0.07647191129018724}

```
for t in text:
    text1[t] = (1/(math.sqrt(len(text)))) * (1/(self.numberOfDocuments +1))
```

text length 19

```
['#Edinburgh',
'#cyclinglife',
'another', 'care',
'chilly', 'cycling',
'cyclists', 'felt',
'good', 'like',
'morning', 'one',
'saturday', 'saw',
'stay', 'take', 'today',
'two', 'warm']
```

For each term:

$1 / (\text{SQRT}(19) * (1/3))$

```
'cycling': 0.07647191129018724,
• {'#Edinburgh': 0.07647191129018724, '#cyclinglife':
0.07647191129018724, 'another': 0.07647191129018724,
'care': 0.07647191129018724, 'chilly':
0.07647191129018724, 'cycling': 0.07647191129018724,
'cyclists': 0.07647191129018724, 'felt':
0.07647191129018724, 'good': 0.07647191129018724,
'like': 0.07647191129018724, 'morning':
0.07647191129018724, 'one': 0.07647191129018724,
'saturday': 0.07647191129018724, 'saw':
0.07647191129018724, 'stay': 0.07647191129018724,
'take': 0.07647191129018724, 'today':
0.07647191129018724, 'two': 0.07647191129018724, 'warm':
0.07647191129018724}
```

Reweighted vectors

- {'blacks': 0.13900960937138315, 'car': 0.13900960937138315, 'chats': 0.13900960937138315, 'driver': 0.13900960937138315, 'east': 0.13900960937138315, 'genuine': 0.13900960937138315, 'guys': 0.13900960937138315, 'knows': 0.13900960937138315, 'london': 0.11350087076783312, 'proper': 0.13900960937138315, 'someone': 0.13900960937138315, 'uber': 0.13900960937138315, '335pm': 0.11350087076783312, '@SW_Help': 0.11350087076783312, 'alternative': 0.11350087076783312, 'booked': 0.11350087076783312, 'giving': 0.11350087076783312, 'home': 0.11350087076783312, 'landslip': 0.11350087076783312, 'like': 0.18997278205802, 'looks': 0.11350087076783312, 'planner': 0.11350087076783312, 'running': 0.11350087076783312, 'see': 0.11350087076783312, 'something': 0.11350087076783312, 'southampton': 0.11350087076783312, 'today': 0.11350087076783312, 'travel': 0.11350087076783312, 'waterloo': 0.11350087076783312}
- {'#Edinburgh': 0.07647191129018724, '#cyclinglife': 0.07647191129018724, 'another': 0.07647191129018724, 'care': 0.07647191129018724, 'chilly': 0.07647191129018724, 'cycling': 0.07647191129018724, 'cyclists': 0.07647191129018724, 'felt': 0.07647191129018724, 'good': 0.07647191129018724, 'like': 0.07647191129018724, 'morning': 0.07647191129018724, 'one': 0.07647191129018724, 'saturday': 0.07647191129018724, 'saw': 0.07647191129018724, 'stay': 0.07647191129018724, 'take': 0.07647191129018724, 'today': 0.07647191129018724, 'two': 0.07647191129018724, 'warm': 0.07647191129018724}

Grouping tweets

- {'changing': 0.373665070649373, 'https://t.co/9ekXqXLUeq': 0.019748812638334105, 'invited': 0.373665070649373, 'keep': 0.373665070649373, 'landlords': 0.373665070649373, 'rules': 0.373665070649373, 'webinar': 0.373665070649373, 'https://t.co/siO3uXr0xE': 0.05225044692984142, 'https://t.co/tR7Q7VWf5e': 0.1396451933694864, 'https://t.co/eB0b8EDvZM': 0.373665070649373}
- 1351844052941959170
- 1351844052950425601
- 1351844053923491840
- 1351844054317686786
- ['changing', 'https://t.co/9ekXqXLUeq', 'invited', 'keep', 'landlords', 'rules', 'webinar']
- ['changing', 'https://t.co/siO3uXr0xE', 'invited', 'keep', 'landlords', 'rules', 'webinar']
- ['changing', 'https://t.co/tR7Q7VWf5e', 'invited', 'keep', 'landlords', 'rules', 'webinar']
- ['changing', 'https://t.co/eB0b8EDvZM', 'invited', 'keep', 'landlords', 'rules', 'webinar']

Landlords invited to webinar to keep up with changing rules

 Noah Vickers



Published: 10:44 AM January 20, 2021



Alison Webb, executive member of housing, health and environment at Breckland Council. Picture: Keith Mindham - Credit: Photo: Keith Mindham Photography

Landlords and letting agents with properties in Breckland have been invited to attend an online forum to find out the latest news about legislation, grants and housing safety updates.



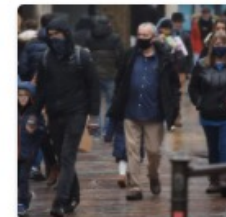
'I will not resign,' says councillor who broke Covid isolation

ABIGAIL NICHOLSON



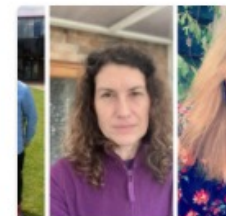
Seven lockdown rules that could change

DANIEL MOXON



Dereham, Blofield and Belton are the latest coronavirus infection hotspots

SAM FERGUSON



'It's opened my eyes' - What is it really like having coronavirus?

DANIEL HICKEY



Four tweets!



Regional Interest in the topic

Tweetids

1351844052941959170
1351844052950425601
1351844053923491840
1351844054317686786

- 2023-01-18 06:02:41
- 2023-01-18 06:09:24

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[('spaces', 0.37780131723056565), ('women', 0.37780131723056565), ('drakeford', 0.21868477295028663),
 ('happy', 0.21868477295028663), ('identify', 0.21868477295028663), ('occupy', 0.21868477295028663),
 ('rapist', 0.21868477295028663), ('self', 0.21868477295028663), ('son', 0.21868477295028663),
 ('6pm', 0.159116544280279), ('@frusinatebhoy', 0.159116544280279), ('apparently', 0.159116544280279),
 ('children', 0.159116544280279), ('crackpot', 0.159116544280279), ('curfew', 0.159116544280279),
 ('dangerous', 0.159116544280279), ('deviant', 0.159116544280279), ('disguised', 0.159116544280279),
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 0.159116544280279), ('men', 0.159116544280279), ('wanted',
 0.159116544280279)] ***** the tweets in the cluster *****
```

```
1 - 1615590481412358144 macaroni1888 :-- : @frusinatebhoy This crackpot wanted
- genuinely wanted - to curfew all males after 6pm.
- But is apparently fine with dangerous deviant men disguised as women entering spaces for women and
  children.
- https://t.co/IhRlM0cQ5D qScore : 0.571474358974359 nScore : 1.7214645838386302
2 - 1615592172010745856 LGB_1962 :-- : Drakeford happy for his rapist son to
  self identify so he can occupy women only spaces. qScore : 0.6081666666666666 nScore
  : 1.829042251701629
```

Top terms

- spaces 0.37780131723056565
- women 0.37780131723056565
- drakeford 0.21868477295028663
- happy 0.21868477295028663
- identify 0.21868477295028663
- occupy 0.21868477295028663
- rapist 0.21868477295028663
- self 0.21868477295028663
- son 0.21868477295028663
- 6pm 0.159116544280279
- @frusinatebhoy 0.159116544280279
- apparently 0.159116544280279
- children 0.159116544280279
- crackpot 0.159116544280279
- curfew 0.159116544280279
- dangerous 0.159116544280279

- [('denzel', 0.31507569531533086), ('dumfries', 0.31507569531533086), ('inter', 0.31507569531533086), ('ahead', 0.2171929849239275), ('building', 0.2171929849239275), ('clash', 0.2171929849239275), ('italiana', 0.2171929849239275), ('milan', 0.2171929849239275), ('photo', 0.2171929849239275), ('supercoppa', 0.2171929849239275), ('wednesday', 0.2171929849239275), ('wingback', 0.2171929849239275), ('#ForzaInter', 0.11988134754977947), ('#Internazionale', 0.11988134754977947), ('#Internisti', 0.11988134754977947), ('#Nerazzuri', 0.11988134754977947), ('#COYS', 0.0978827103914034), ('#THFC', 0.0978827103914034), ('23yearold', 0.0978827103914034), ('26yearold', 0.0978827103914034), ('@SempreIntercom', 0.0978827103914034), ('antonio', 0.0978827103914034), ('better', 0.0978827103914034), ('characteristics', 0.0978827103914034), ('conte', 0.0978827103914034), ('decided', 0.0978827103914034), ('interested', 0.0978827103914034), ('pedro', 0.0978827103914034), ('player', 0.0978827103914034), ('porro', 0.0978827103914034), ('prefer', 0.0978827103914034), ('rightback', 0.0978827103914034), ('sporting', 0.0978827103914034), ('spurs', 0.0978827103914034), ('technical', 0.0978827103914034), ('tier', 0.0978827103914034), ('wants', 0.0978827103914034), ('#FCIM', 0.09731163737414802), ('#InterMilan', 0.09731163737414802)]
- ***** the tweets in the cluster *****
- 1 - 1615589931195187204 InsideSpurs1 :-- : Spurs are NOT interested in 26-year-old Inter right-back Denzel Dumfries and prefer Sporting s 23-year-old Pedro Porro. Antonio Conte has decided he wants a player with better technical characteristics than Dumfries. @SempreIntercom - Tier #THFC || #COYS https://t.co/aI8NEBz5k8 qScore : 0.6179394977168949 nScore : 4.4073424851295515
- 2 - 1615591069046939650 InterMilanNow :-- : Photo Inter Milan Wingback Denzel Dumfries Ahead Of Supercoppa Italiana Clash Vs AC Milan: Building Up To Wednesday #ForzaInter #Internisti #Nerazzuri #Internazionale https://t.co/j69E8iq094 qScore : 0.6718820861678004 nScore : 1.9407670315888352
- 3 - 1615592490526023680 intermilan :-- : Photo - Inter Milan Wingback Denzel Dumfries Ahead Of Supercoppa Italiana Clash Vs AC Milan: "Building Up To Wednesday" #InterMilan #FCIM https://t.co/DBiPW8BvCN qScore : 0.6643733850129199 nScore : 2.8329581689327106
- -----

Top terms

- 91
- 2023-01-18 06:00:30
- 2023-01-18 06:10:40
- 3
- 3
- -2.38888888888889
- denzel 0.31507569531533086
- dumfries 0.31507569531533086
- inter 0.31507569531533086
- ahead 0.2171929849239275
- building 0.2171929849239275
- clash 0.2171929849239275
- italiana 0.2171929849239275
- milan 0.2171929849239275
- photo 0.2171929849239275
- supercoppa 0.2171929849239275
- wednesday 0.2171929849239275

- wingback 0.2171929849239275
- #ForzaInter 0.11988134754977947
- #Internazionale 0.11988134754977947
- #Internisti 0.11988134754977947
- #Nerazzuri 0.11988134754977947
- #COYS 0.0978827103914034
- #THFC 0.0978827103914034
- 23yearold 0.0978827103914034
- 26yearold 0.0978827103914034
- @SempreIntercom 0.0978827103914034
- antonio 0.0978827103914034
- better 0.0978827103914034
- characteristics 0.0978827103914034
- conte 0.0978827103914034
- decided 0.0978827103914034
- interested 0.0978827103914034
- pedro 0.0978827103914034
- player 0.0978827103914034
- porro 0.0978827103914034
- prefer 0.0978827103914034
- rightback 0.0978827103914034
- sporting 0.0978827103914034
- spurs 0.0978827103914034
- technical 0.0978827103914034
- tier 0.0978827103914034